

Plant-Water Relations as Drivers of Plant Function and Ecology

Dissertation Proposal

Juan Manuel Cely Arévalo - PhD Student

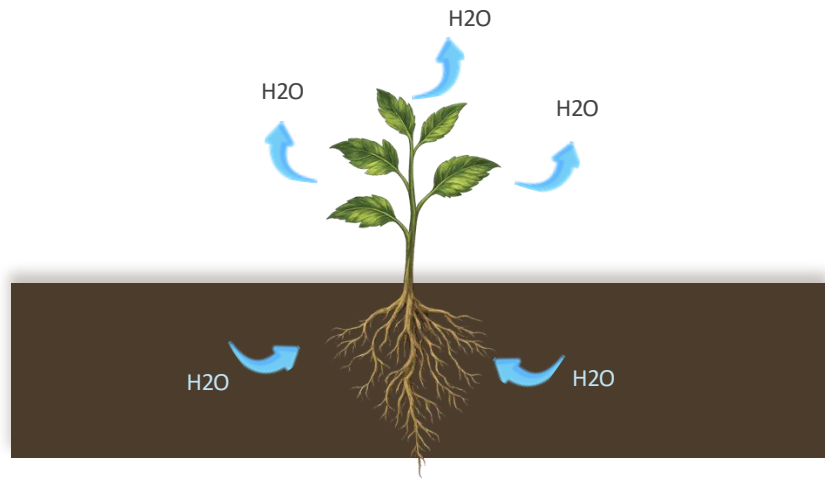
Advised by Dr. María Natalia Umaña

M | ECOLOGY AND EVOLUTIONARY BIOLOGY



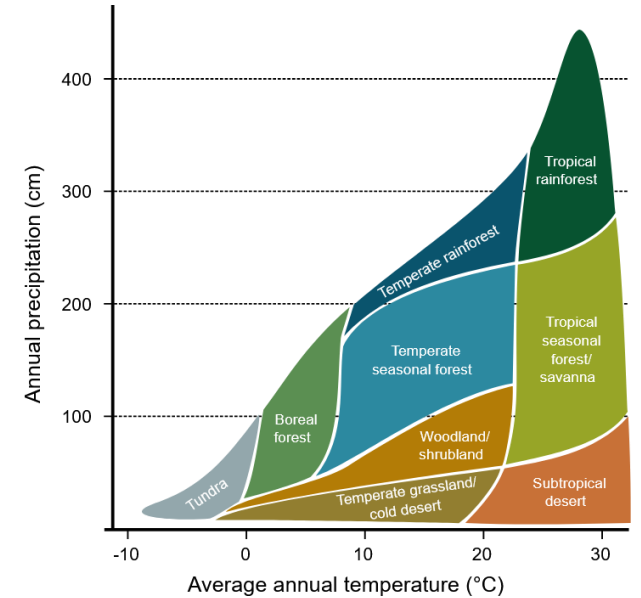
INTRODUCTION

Plants need water to live



Water is expected to shape plants strategies and distribution

Water availability determines biome distribution.

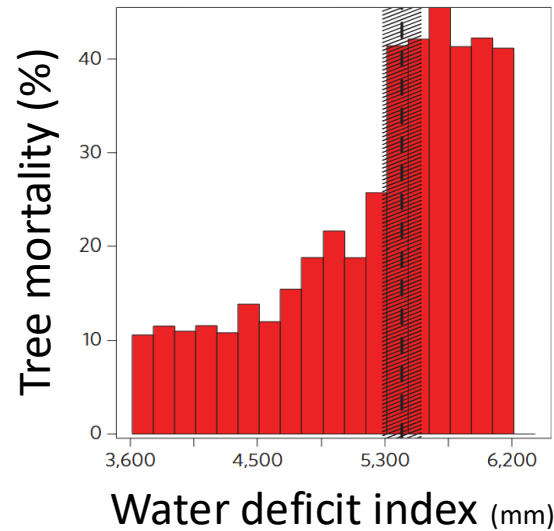


+ Drought tolerance

+ Water use

WHY TO TOLERATE DROUGHT?

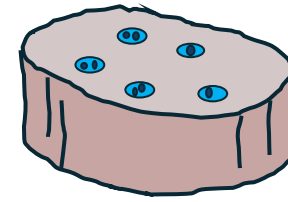
Drought strongly influence tree mortality



Anderegg et al. (2015)

Strategies that avoid this damage will allow plants to thrive under drought

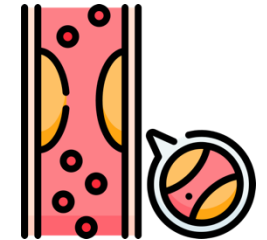
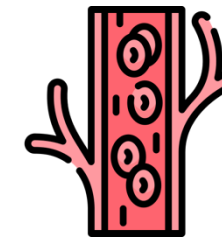
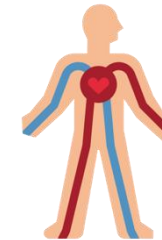
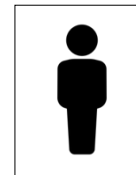
Drought damage plant vessels



Functional vessel



Vessel with embolism



Zimmerman (1989); Sperry et al. (2007)

TROPICAL DRY FORESTS



DRY FORESTS ADAPTATIONS TO DROUGHT

Plants in these forests have found strategies to cope with drought.

Trees here drop their leaves too!



Tropical Dry Forest in
Guanacaste, Costa Rica, 2024



Ceiba sp.
Santa Fe, Colombia, 2024



Bursera simaruba
Source: Inst. Humboldt

Spiky stems that
can do
photosynthesis
:s

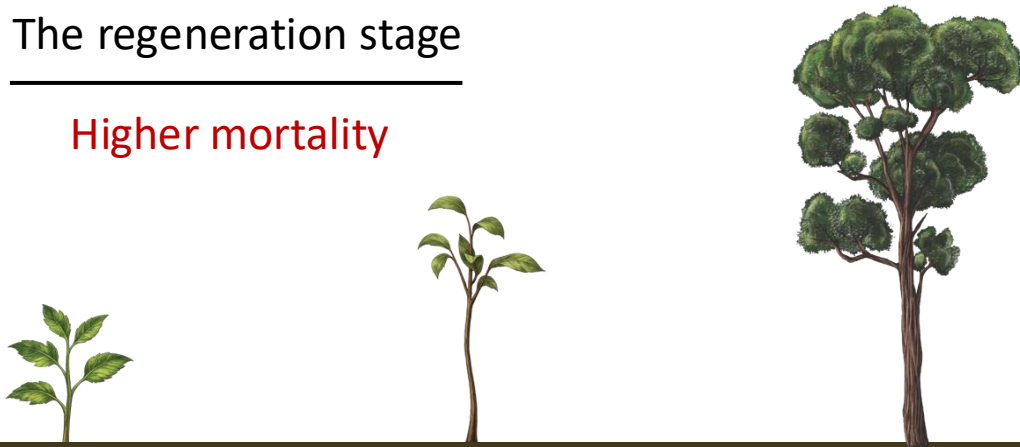
TROPICAL DRY FORESTS

Important progress has been made in understanding these forests.

...however, key gaps remain.

The regeneration stage

Higher mortality



Conditions across lifetime differ

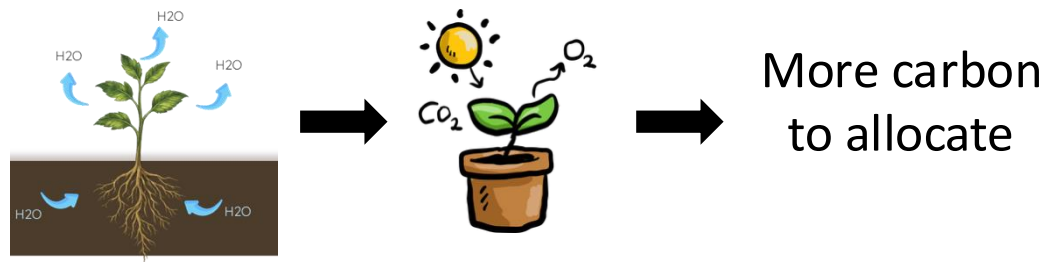
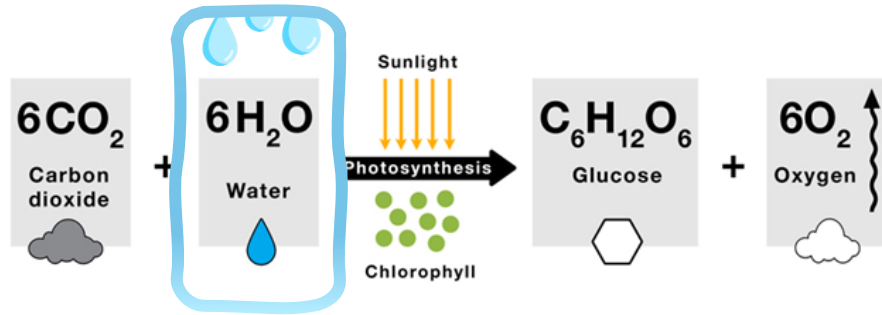
The way seedlings manage water might be key for species success

- 1 The diversity of strategies that plants adopt at this stage
- 2 The way these strategies determine forest composition

Condit (1995); Barton (2024); González-M (2021)

WATER ALSO ENHANCE CARBON FIXATION

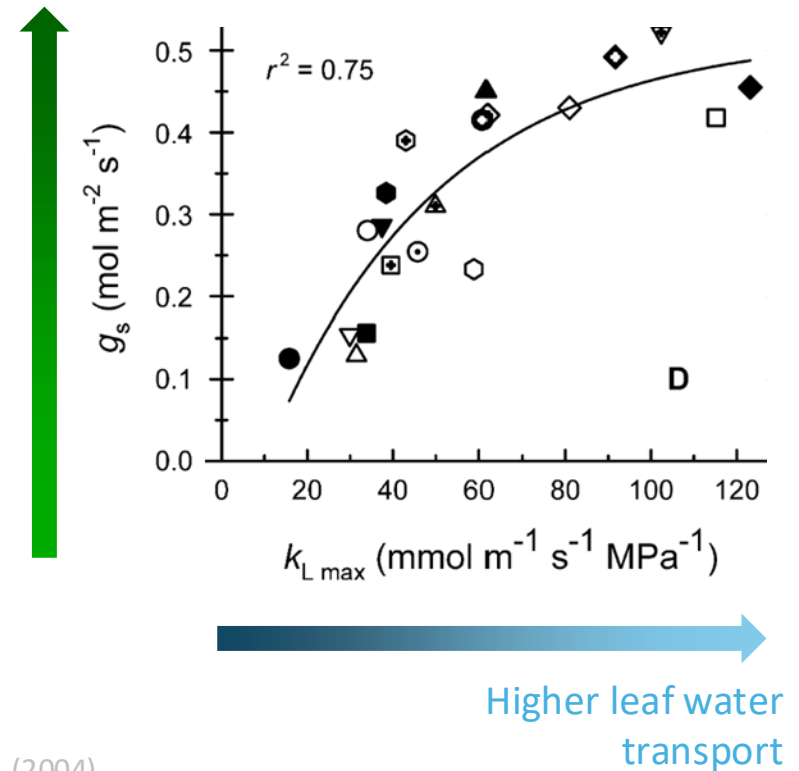
Not everything is about mortality...
water is life.



Water-use directly determines
how much carbon plants have to invest

Higher water transport capacity
increases photosynthesis

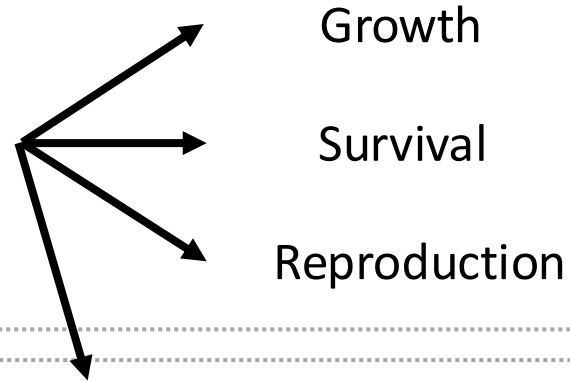
Higher
photosynthesis



Santiago et al. (2004)

TROPICAL DRY FORESTS

How do plants
invest their
fixed carbon?



DEFENSE

is one of these key functions

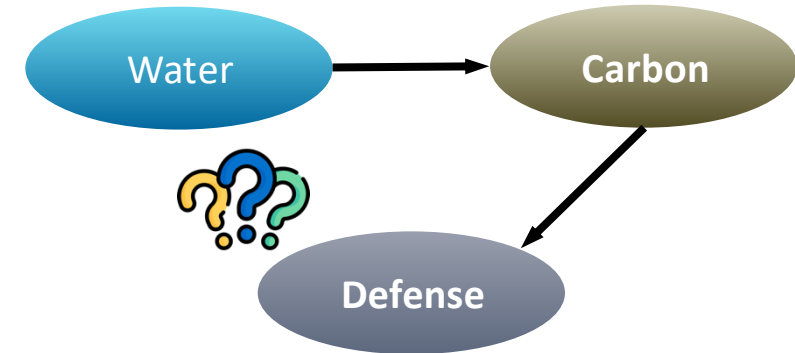


Extrafloral nectaries give
away carbon



In exchange, they receive
defense.

Bentley (1977)



A lot of research in each of these
processes independently

Coley (1989); Herms & Mattson (1992); Reich (1998); Wright et al. (2004); Choat et al. (2008); Reich (2014); Gleason et al. (2016); Oliveira et al. (2021); Umaña et al. (2023); Endara et al. (2023); Glos et al. (2026)

...but no clear link between them

- 3 The effect of defense on water-use strategies

GOALS OF THIS DISSERTATION

Overarching goal

This dissertation aims to explore

How water-use strategies shape plant function and ecology

Questions

- 1 What water-use strategies can be found in these forests?
- 2 How do these strategies determine forest composition?
- 3 What are the links between leaf water-use and plant defense?

Scale

Tropical Dry Forest in Colombia



Global analysis



TROPICAL DRY FORESTS IN THE NEOTROPICS



Dryflor (2016)

WATER

Tropical forests that undergo 3–6 months of drought each year.

SOIL

Rich soils that has been one reason for deforestation.

CONSERVATION

Less than 10% of the original cover remains.

One of the most threatened ecosystems on the planet.

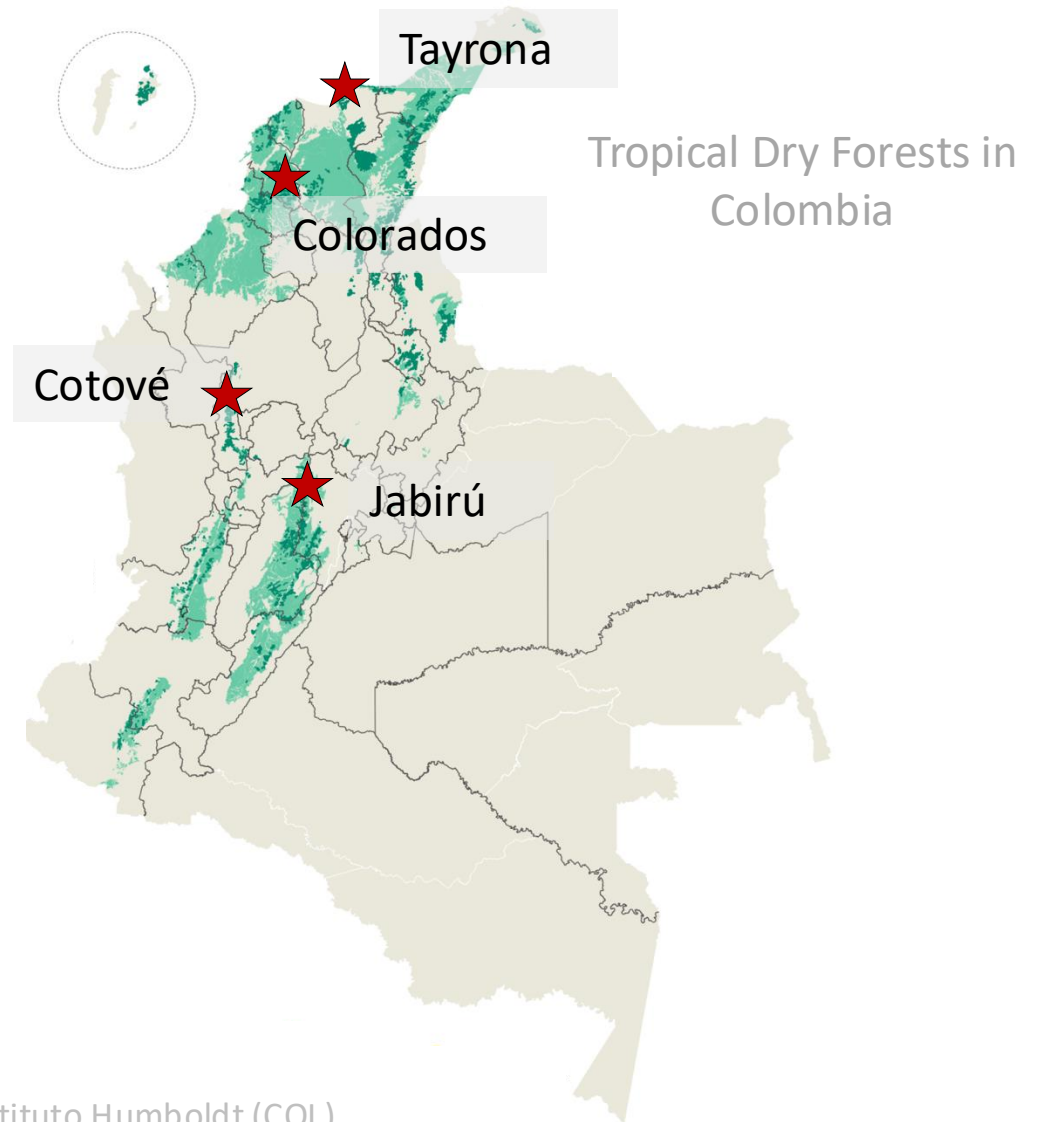
Dryflor (2016); Pizano et al. (2017); Pennington et al. (2018); González-M (2017)

TROPICAL DRY FORESTS IN COLOMBIA

Colombia is a tropical country in the north of South America



source: Wikipedia



source: Instituto Humboldt (COL)

TROPICAL DRY FORESTS

PLANTS

Around 2,500 plant species

ANIMALS

60 mammal species

~230 bird species

High amphibian endemism



Saguinus oedipus



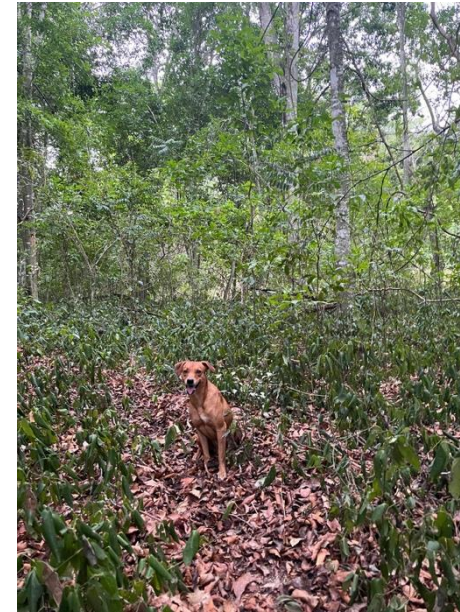
TROPICAL DRY FORESTS IN COLOMBIA



July 2021



July 2024



May 2024



January 2024



Thank you,
Tomás Fuentes-Rowhrer!

Chapter 1

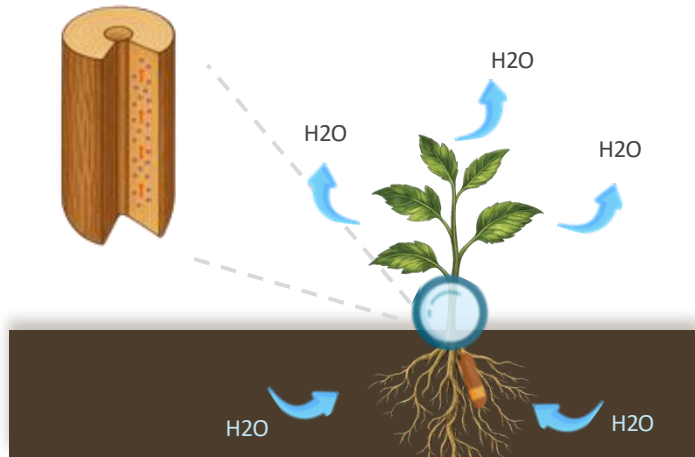
What plant–water relationships occur in tropical dry forests?



Cuniculus paca

XYLEM IS KEY FOR WATER TRANSPORT

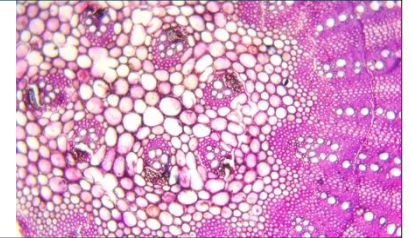
The amount of water that flows through the plant is key for their function



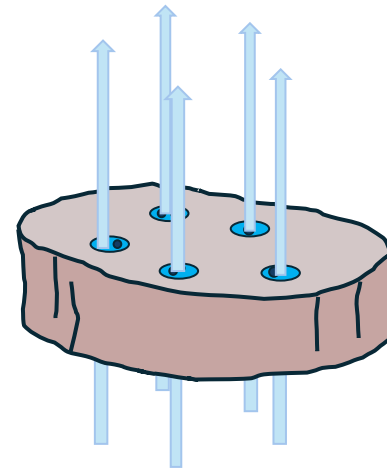
The xylem represents up to 95% of the total pathway that water travels.

Tyree & Zimmerman (2002); Cruiziat (2002)

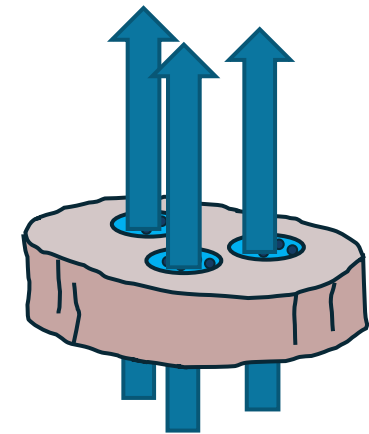
Xylem is the collection of tissues responsible for transporting water.



Vessels are perhaps the most important tissue determining water transport.



Smaller vessels



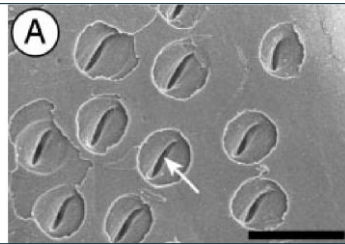
Larger vessels

Sperry et al. (2011)

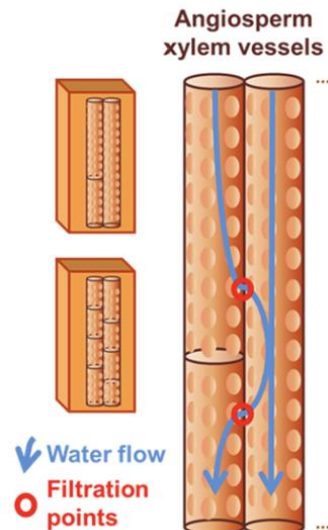
XYLEM IS KEY FOR WATER TRANSPORT

Water can also move “horizontally”

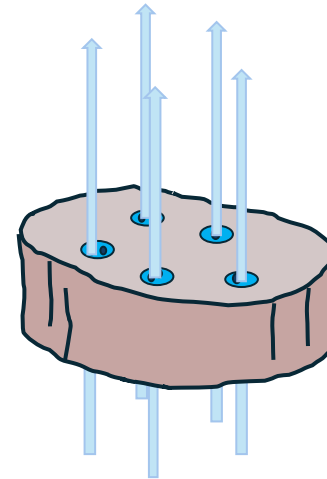
Pit membranes are porous membranes between adjacent vessels



Jansen et al. (2004)

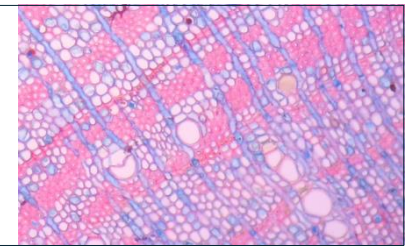


Both vessel traits can enhance drought tolerance



Smaller vessels are less likely to suffer from embolism

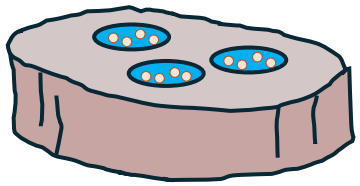
Embolism is the blockage of xylem vessels by an air bubble.



Gleason et al. (2016)

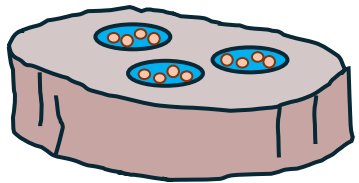
XYLEM IS KEY FOR WATER TRANSPORT

Both traits can enhance drought tolerance



Tolerant

Smaller pit membranes block embolism from spreading.



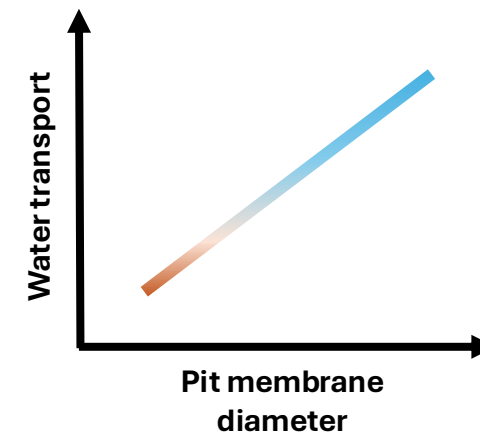
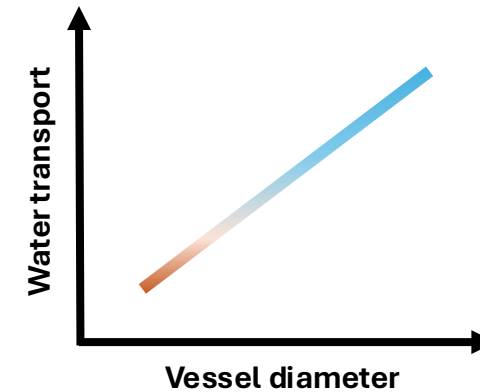
Vulnerable

Larger pits allow higher horizontal flow



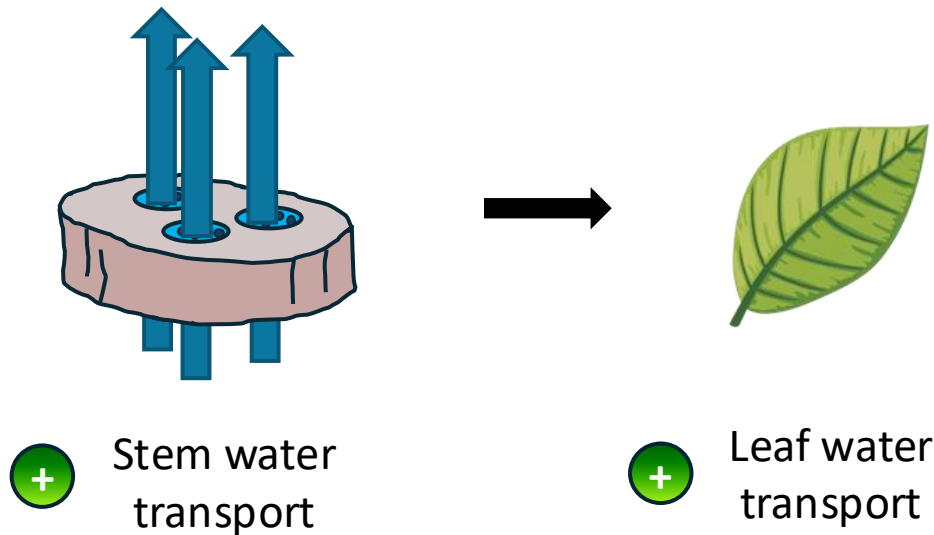
Choat et al. (2008)

Xylem traits should determine water transport



WATER TRANSPORT WILL SHAPE OTHER ORGANS STRATEGIES

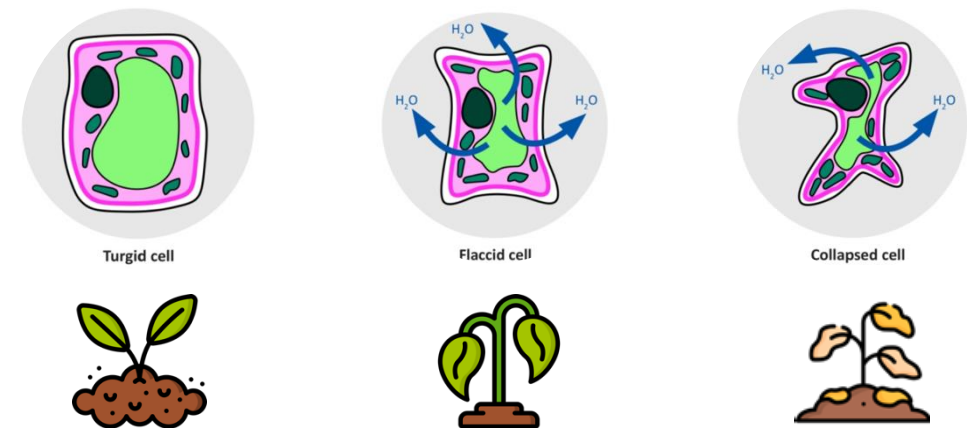
Stem water transport may also shape leaf water-use



Leaf function should also correspond to its water transport capacity

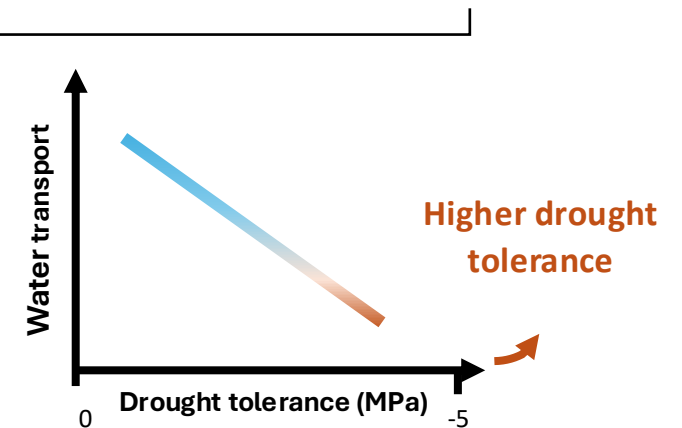
Santiago et al. (2008); Reich (2014)

How to measure leaf drought tolerance?



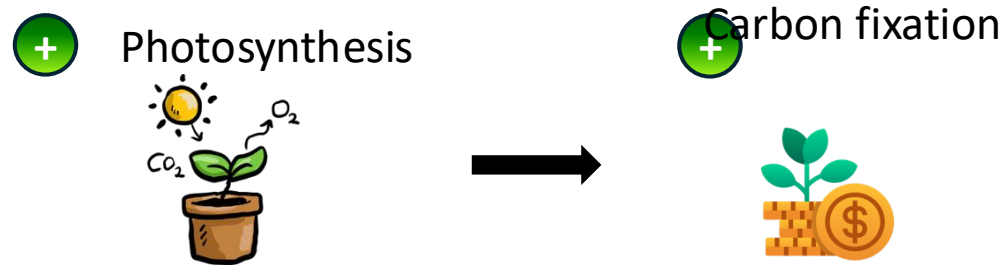
Leaf Turgor Loss Point (MPa)

Bartlett et al. (2012)



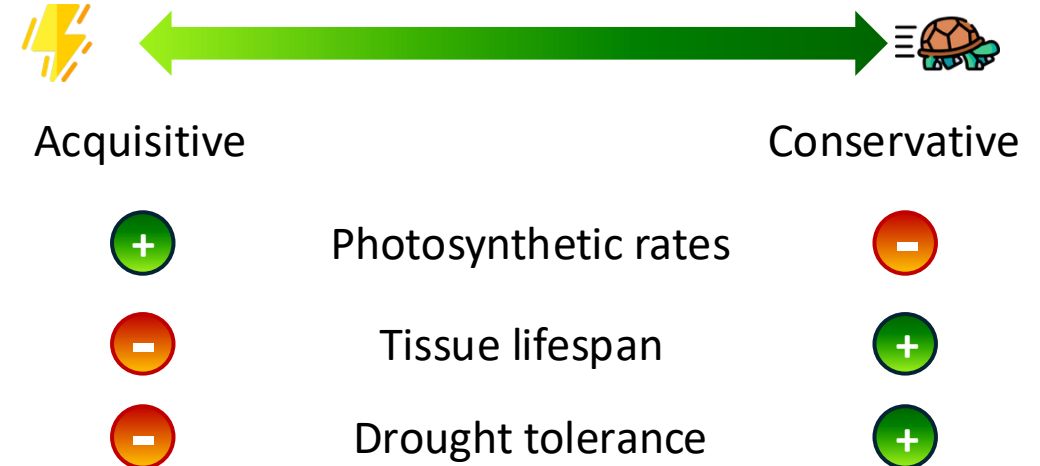
WATER TRANSPORT SHAPES CARBON-ECONOMICS

More efficient **water transport** will support higher carbon fixation rates

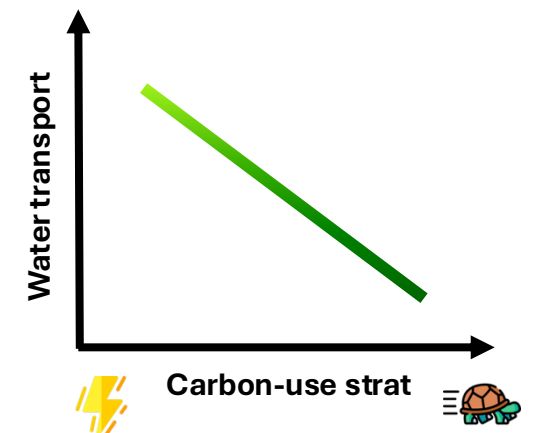


A large body of literature studies the variation of carbon economics across taxa.

Reich (1998); Wright et al. (2004); Sack & Scoffoni (2013); Reich (2014); Oliveira (2021)



Higher water transport should correspond to acquisitive strategies



Cochlospermum orinocense



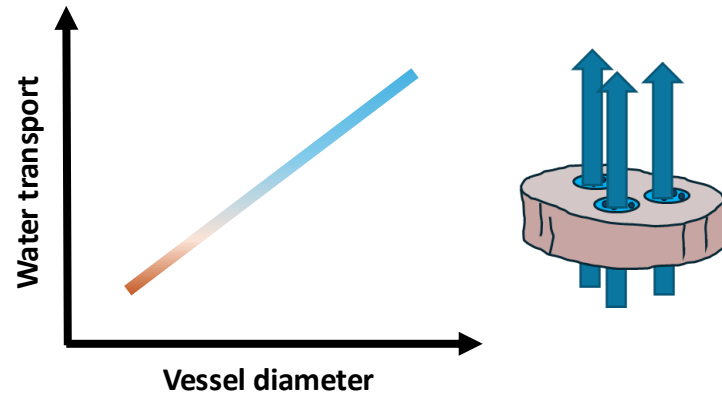
How does **water transport** shape plant functional strategies in tropical dry forests?



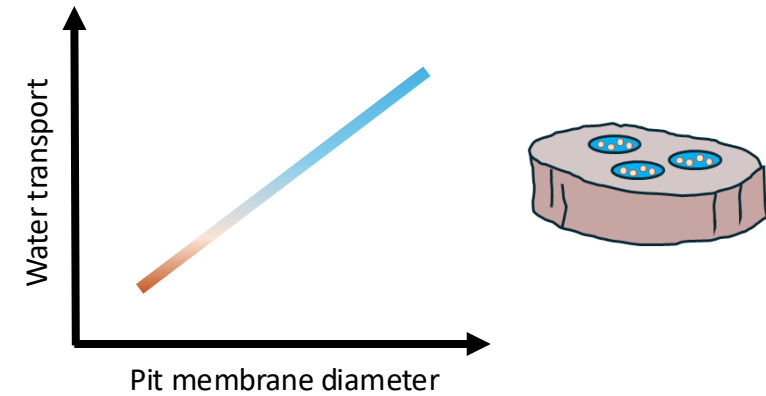
Albizia saman

HOW DOES WATER TRANSPORT SHAPE PLANT–WATER STRATEGIES IN TROPICAL DRY FORESTS?

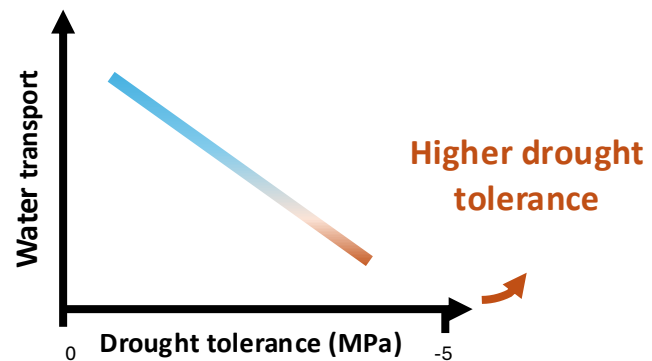
Vessel diameter



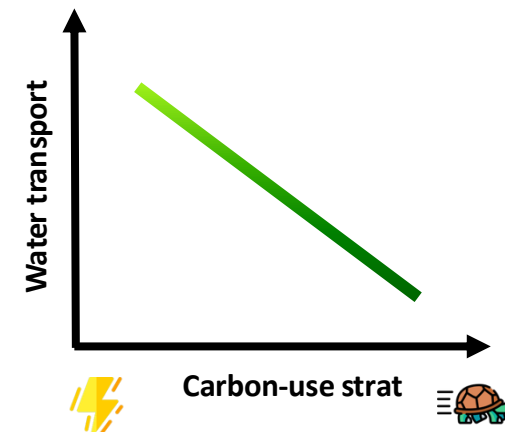
Pit membrane diameter



Leaf drought tolerance



Carbon economics



- ✓ 28 tree species
- ✓ 3 Dry Forests

Cotové



Colorados

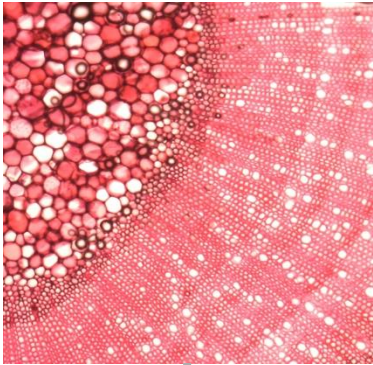


Jabirú



FUNCTIONAL TRAITS AND ANALYSES

Hydraulic traits



Anatomy in the stem



Water transport measured in the stems



Drought tolerance in the leaf

Carbon economics traits



Across the whole plant

+8 continuous months in the field

Measurements of hydraulic conductivity for 140 individuals

A set physiological traits that allow for better assessment of plant function

FUNCTIONAL TRAITS AND ANALYSES

Water
source



A system of hoses
and valves

A balance to
measure water flow

Connect stem to
the system and
measure

FUNCTIONAL TRAITS AND ANALYSES



February 2024

Special thanks to Dr. Beatriz Salgado-Negret, Juliana Puentes, and Felipe Rojas.



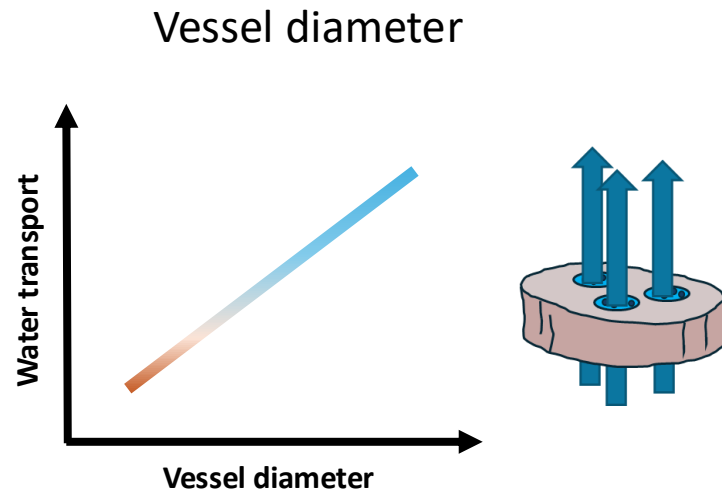
July 2024



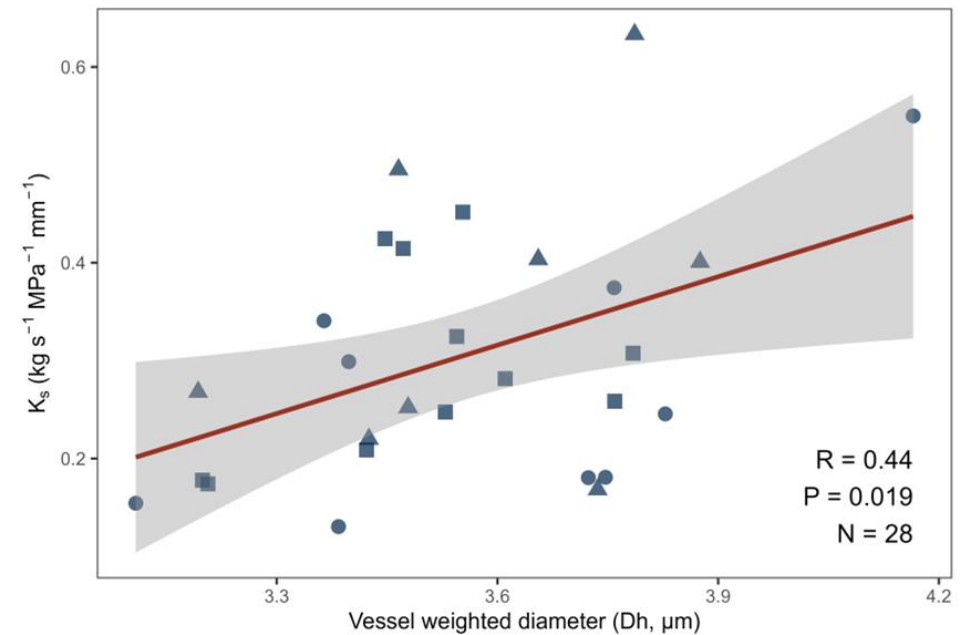
Summer 2025

Special thanks to Dr. Camila Martínez, Sara Vivero, and Valeria Meneses

HOW DOES WATER TRANSPORT SHAPE PLANT–WATER STRATEGIES IN TROPICAL DRY FORESTS?

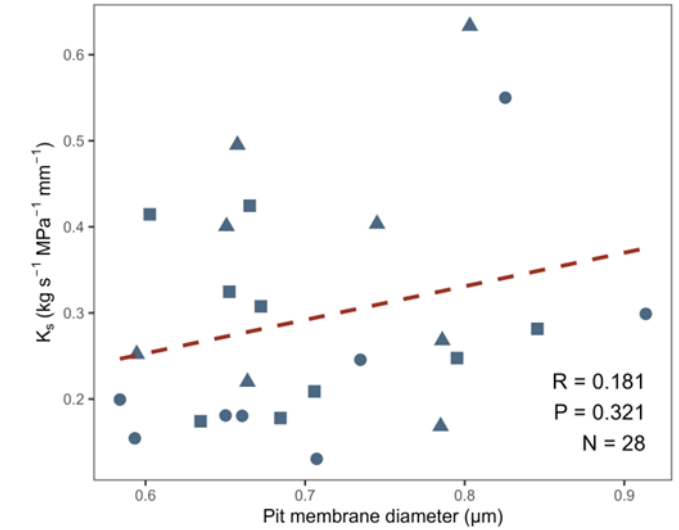
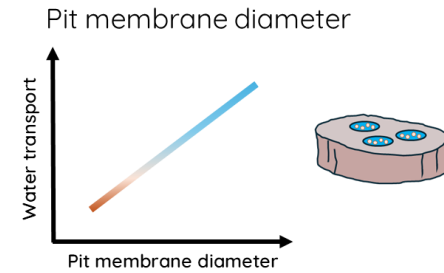


Larger vessel sustain higher
water transport

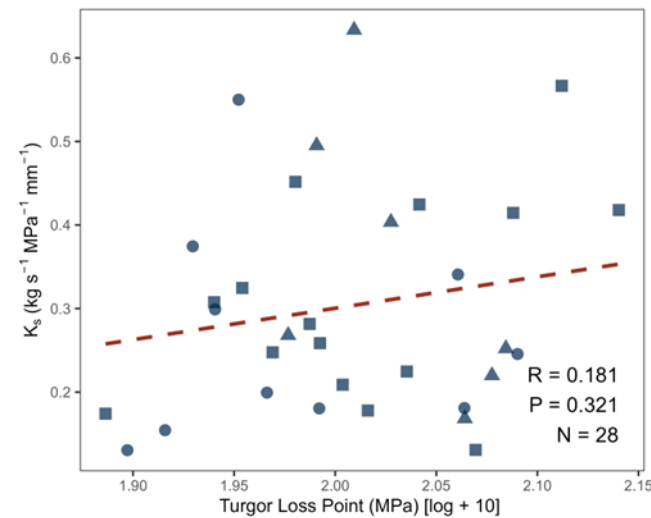
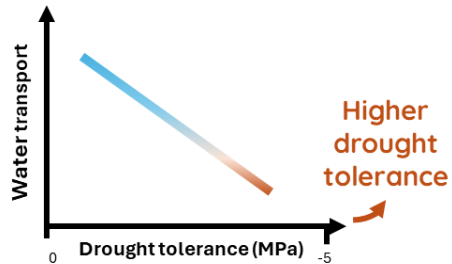


HOW DOES WATER TRANSPORT SHAPE PLANT–WATER STRATEGIES IN TROPICAL DRY FORESTS?

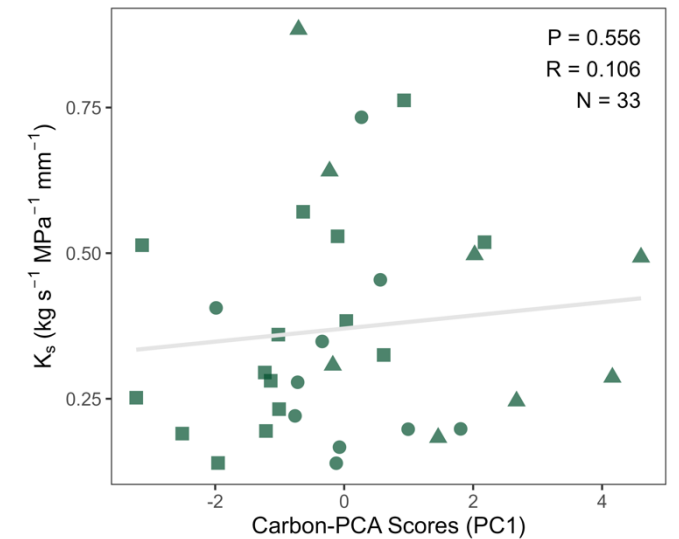
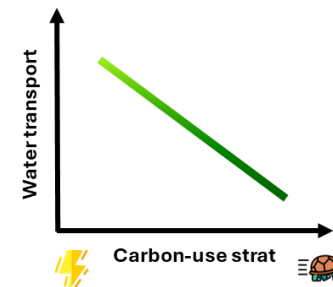
We found no other relationships



Leaf drought tolerance

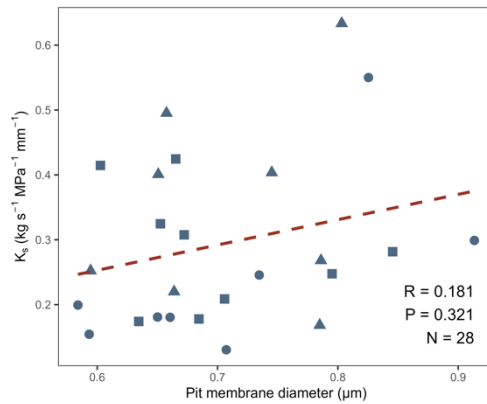
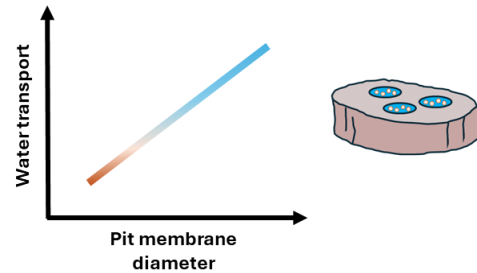


Carbon economics

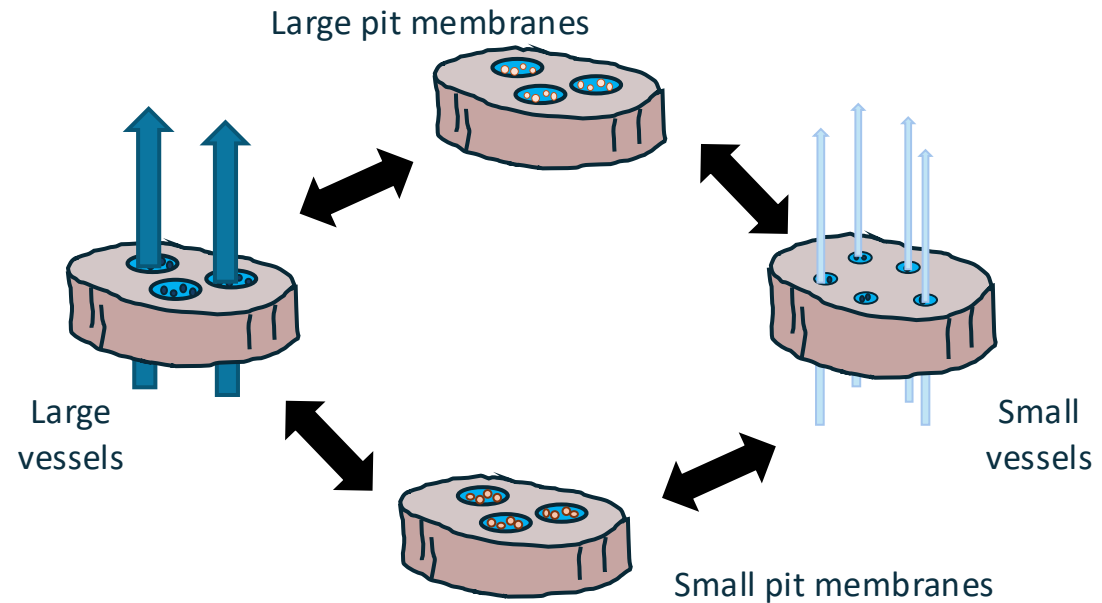


NO BIOPHYSICAL CONSTRAINTS ALONG PLANT FUNCTION

Pit membranes were not related to water transport



Multiple pit membrane–diameter combinations may produce similar function.

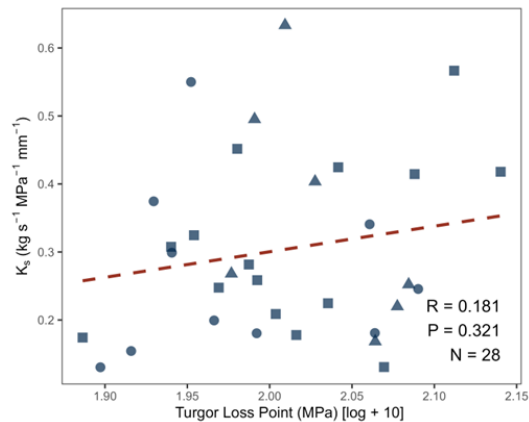
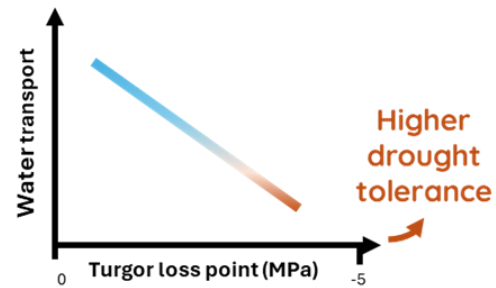


Gleason et al. (2016)

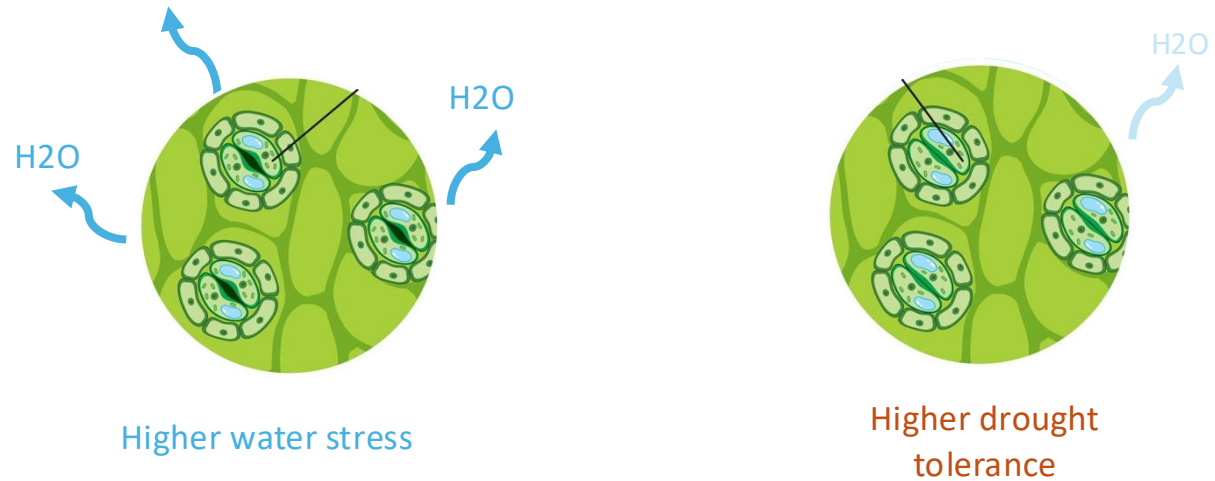
Multiple traits within the vessels determine water transport and drought tolerance

NO BIOPHYSICAL CONSTRAINTS ALONG PLANT FUNCTION

No relation between water transport and leaf tolerance



Effective stomatal regulation can mitigate drought impact.



Mechanisms linking water transport and stomatal regulation are unclear

Cruiziat (2002); Buckley et al. (2019)

Different physiological processes may regulate water-use within the leaf

CONCLUSION

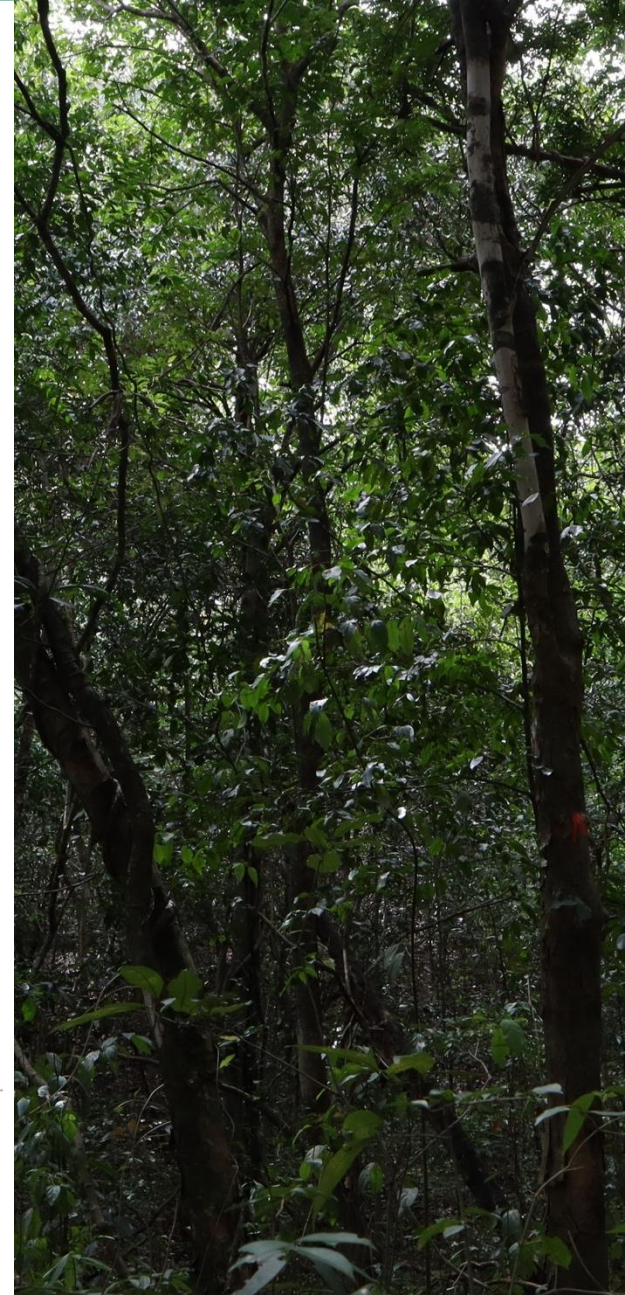


What plant–water relationships occur in tropical dry forests?



Cochlospermum orinocense

Albizia saman



CONCLUSION



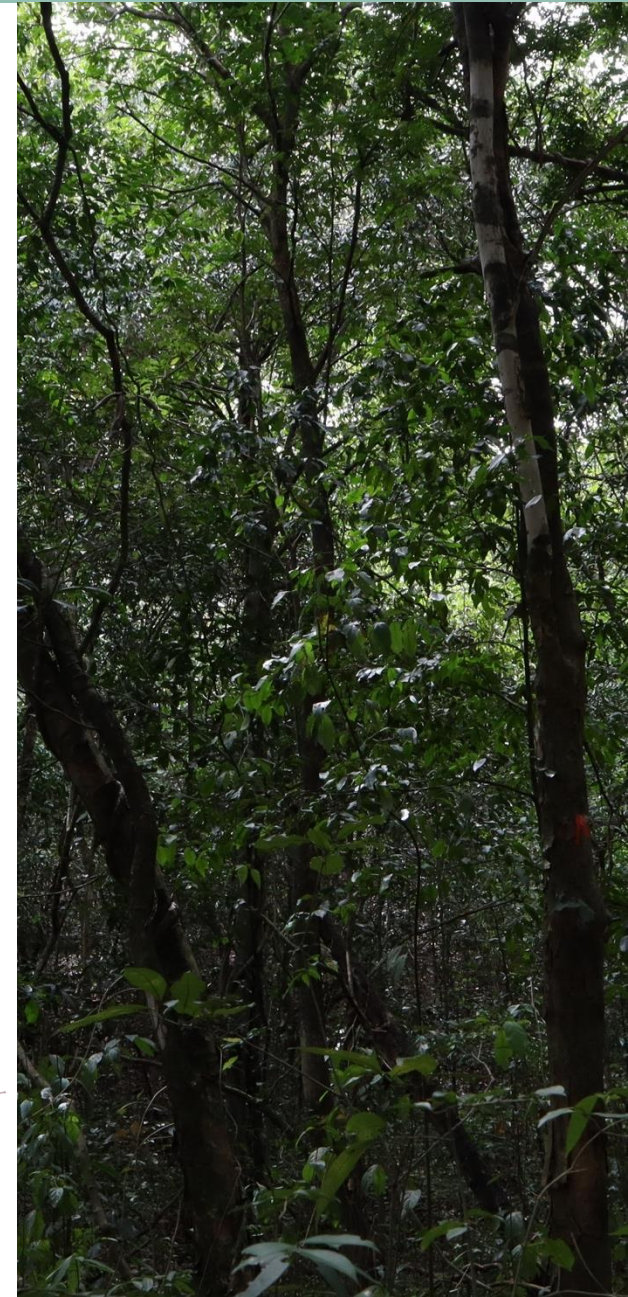
Different combinations of anatomical and physiological traits can shape a complex whole-plant water-use strategy



But wait...

All these species are thriving in the same system, under the same drought stress.

So, if there are so many strategies, are they determining species success at all?



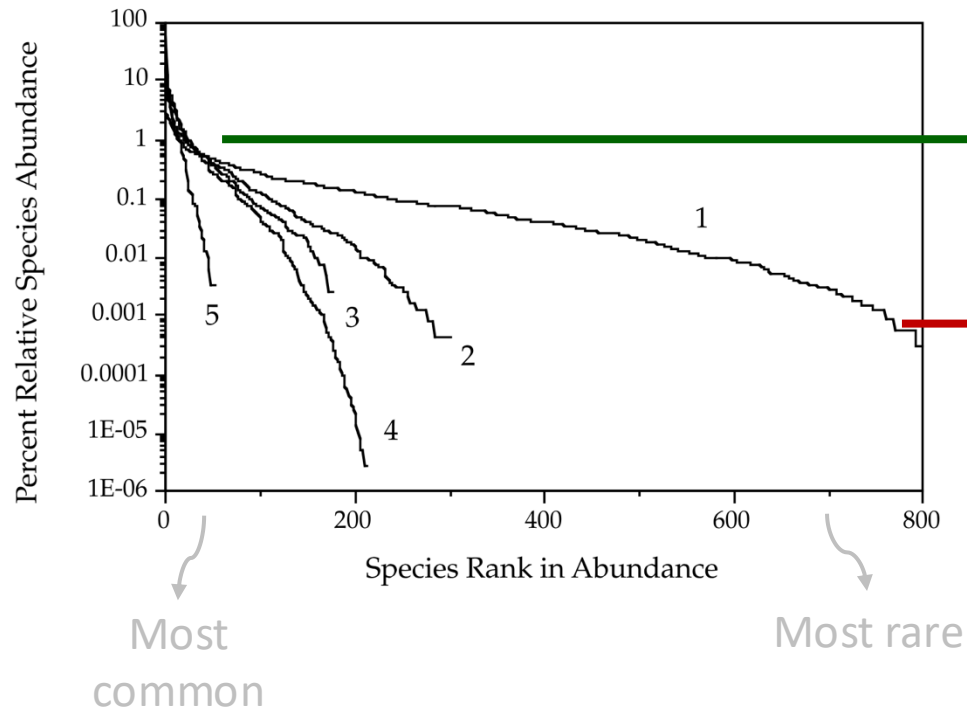
Chapter 2

What's the role of hydraulic strategies determining species abundance?



Leopardus pardalis

WHAT MAKES A SPECIES ABUNDANT?



Few species are very common, while many are rare.

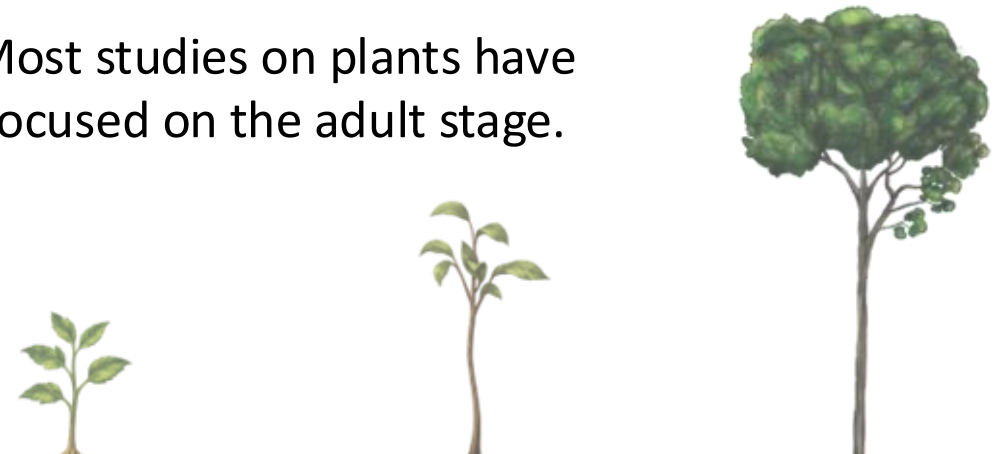
Hubbel (2001)

Classic niche theory offers a clue

Match local conditions = high abundance

No match = low abundance

Most studies on plants have focused on the adult stage.



Conditions across lifetime differ

MacArthur (1965); Barton (2024); Grubb (1977)

SEEDLING STRATEGIES MAY SHAPE SPECIES ABUNDANCES

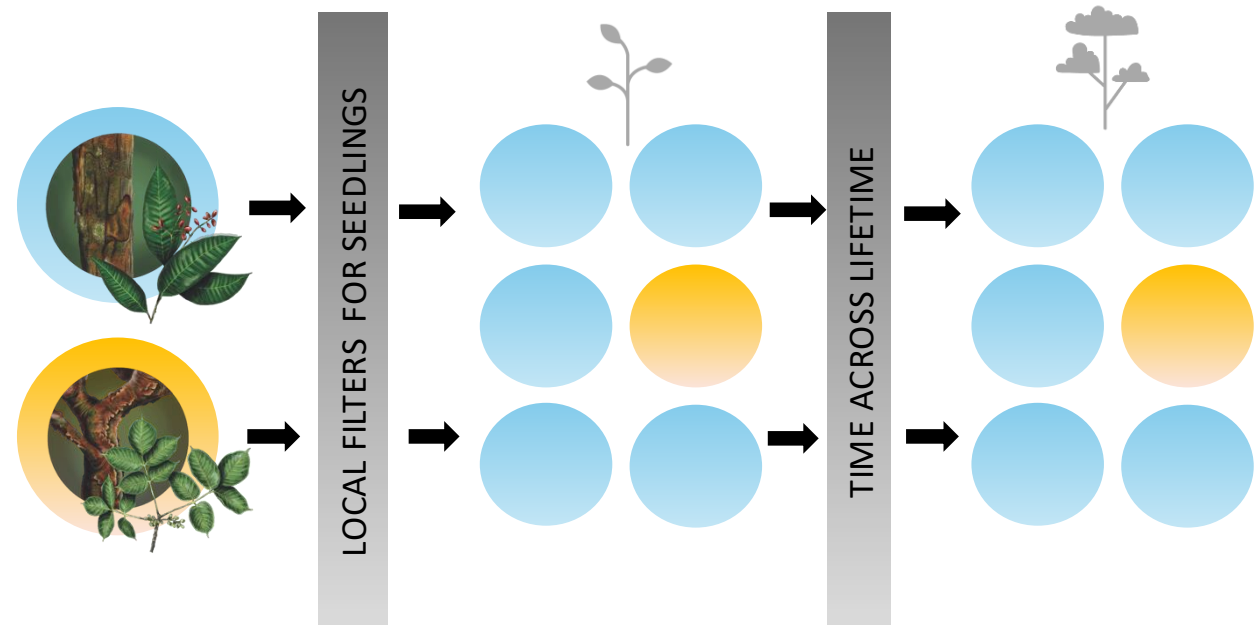
Plant's occupy different
niches through their
lifetime

At the seedling stage...

**The regeneration niche
hypothesis**
provides a framework to
test this gap

Grubb (1977)

Seedling niches may play a key role in
determining species abundance.

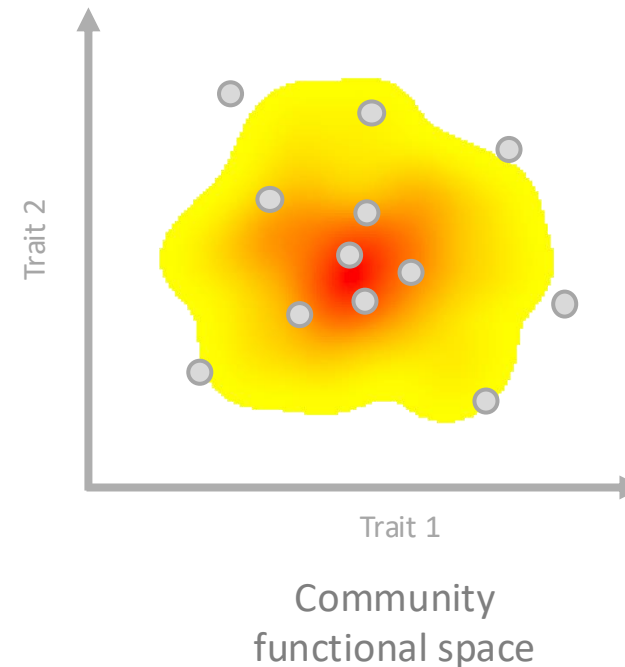


In Chapter I, we found that many strategies are possible.



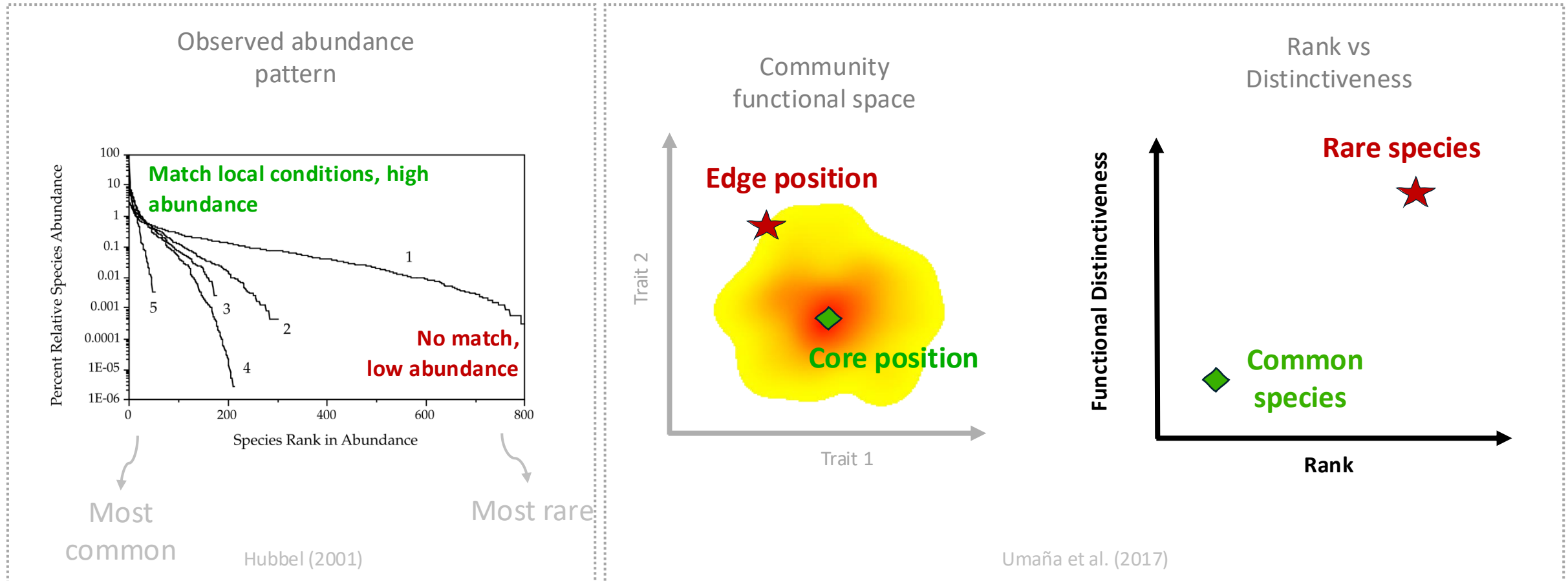
What is the role of water strategies at the seedling stage determining species abundance?

Functional distinctiveness measures how different a species is from the rest of the community.



Violle et al. (2017)

ABUNDANCE MIGHT BE LINKED TO FUNCTIONAL STRATEGIES



QUESTION

Cochlospermum orinocense

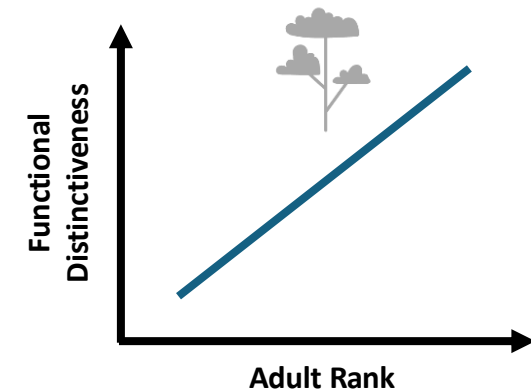
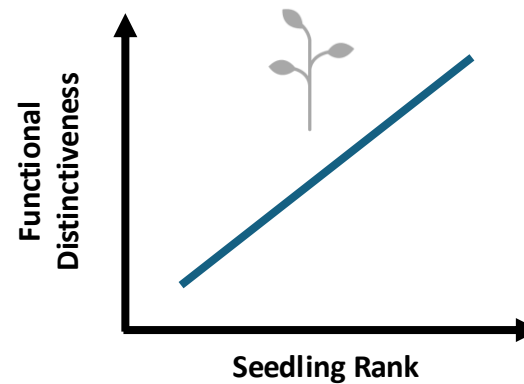


How does seedling hydraulic functional distinctiveness predict abundance across a species' lifetime?



Albizia saman

Under the
**Regeneration Niche
Hypothesis**



Colorados



Jabirú



Cotové



Tayrona



SPECIES INCLUDED IN THIS STUDY (N=48)

15 species

14 species

9 species

11 species

~24% of species richness is represented.

FUNCTIONAL TRAITS AND ANALYSES

Functional traits

7 traits related to whole-plant carbon economics

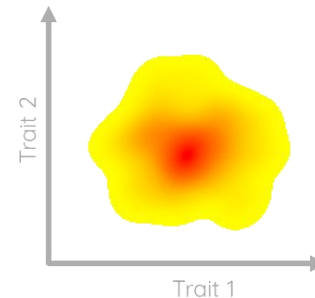
10 wood anatomy traits reflecting water-use strategies



Analyses

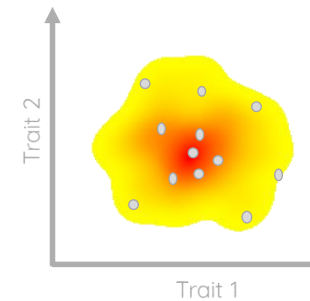
①

Separate PCAs were estimated for carbon-economics and hydraulic strategies.



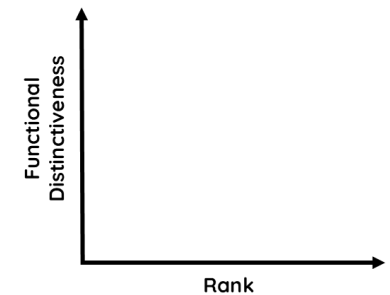
②

Distinctiveness estimated from species' positions in 2D PCA space.



③

Linear models tested the relationship between rank and distinctiveness.



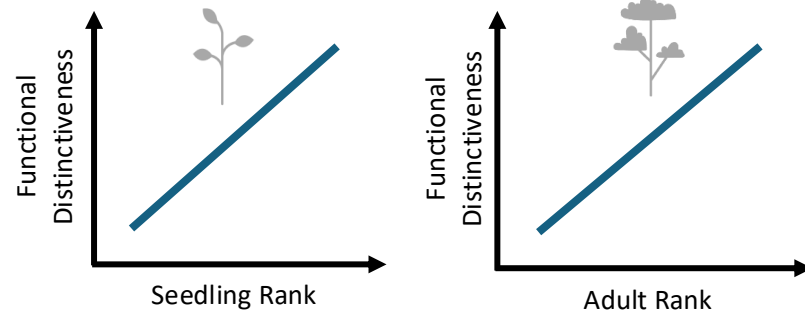
+20 years of
research

More than 5,000
functional trait
measurements

~ 13,000 trees
and seedlings
recorded

PRELIMINARY RESULTS AND DISCUSSION

How does seedling hydraulic functional distinctiveness predict species rarity across ontogeny?

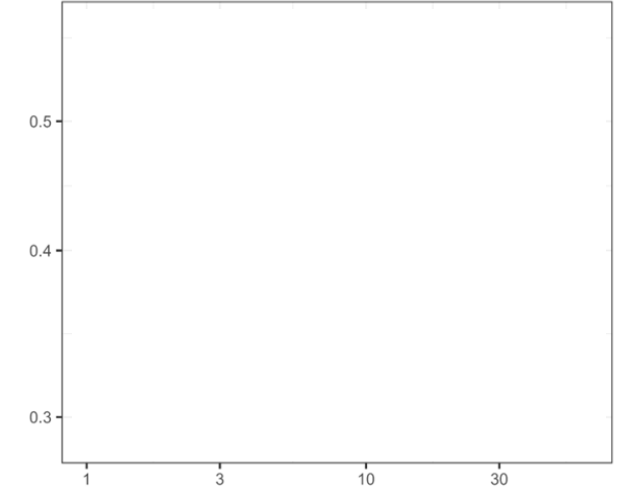
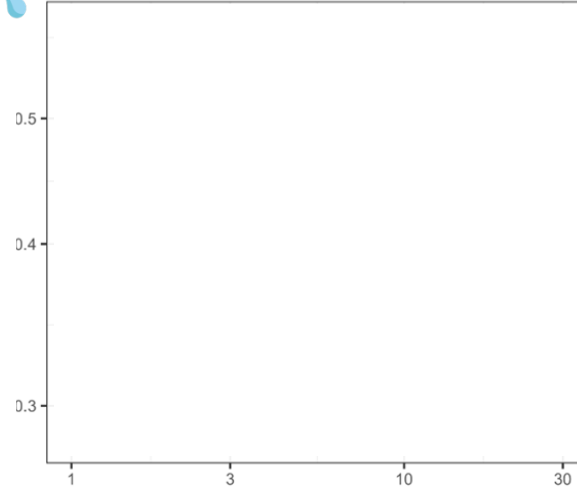


 **Supporting the regeneration niche hypothesis for hydraulic traits**

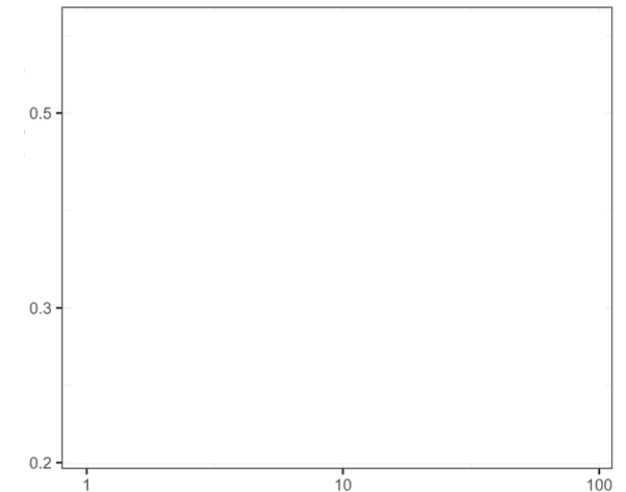
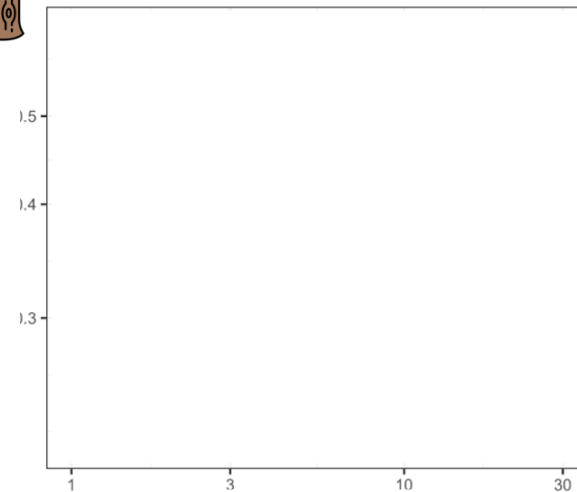
 **First time to empirically link abundance, functional strategies and ontogeny**



Hydraulic distinctiveness



Carbon-use distinctiveness



Rank Seedlings (log)

Rank adults (log)

Cochlospermum orinocense



What's the role of hydraulic strategies
determining species abundance?

Here we show that **optimal hydraulic strategies**
expressed in seedlings
predict species abundance across the lifetime.

Albizia samar





Pheucticus ludovicianus

Chapter 3

How are plant defenses related to plant water-use strategies?

THE IMPORTANCE OF HERBIVORY ON PLANT ECOLOGY

In Chapter II, hydraulic strategies predict a key ecological pattern: species commonness and rarity.

Other key process in ecological studies is the plant–animal interaction of **herbivory**.

PLANT DEFENSE

Each year, around 10% of plant production is consumed by herbivores.

An interaction so ancient that it has shaped plant evolutionary history.



Among many defense strategies, extrafloral nectaries (EFNs) are a very special case.



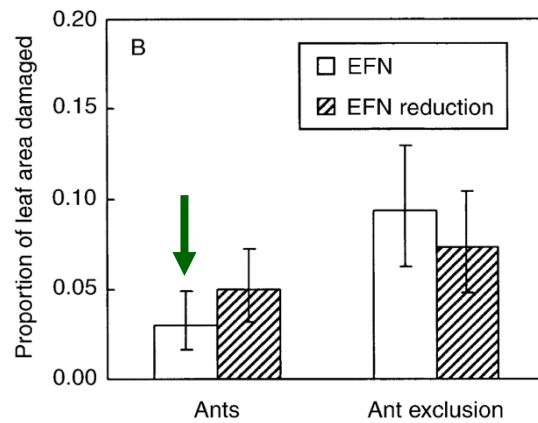
Organs specialized to sustain indirect defenses mediated by bodyguards.



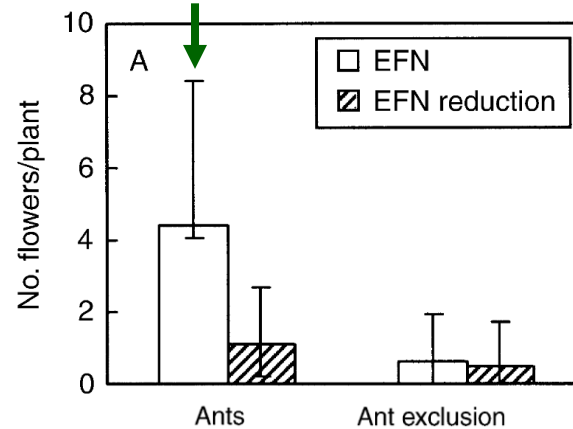
They offer sugar-rich rewards and, in return, receive protection.

A huge investment...
that works.

EXTRAFLOREAL NECTARIES ROLE IN PLANTS



Reduced herbivory
with bodyguards

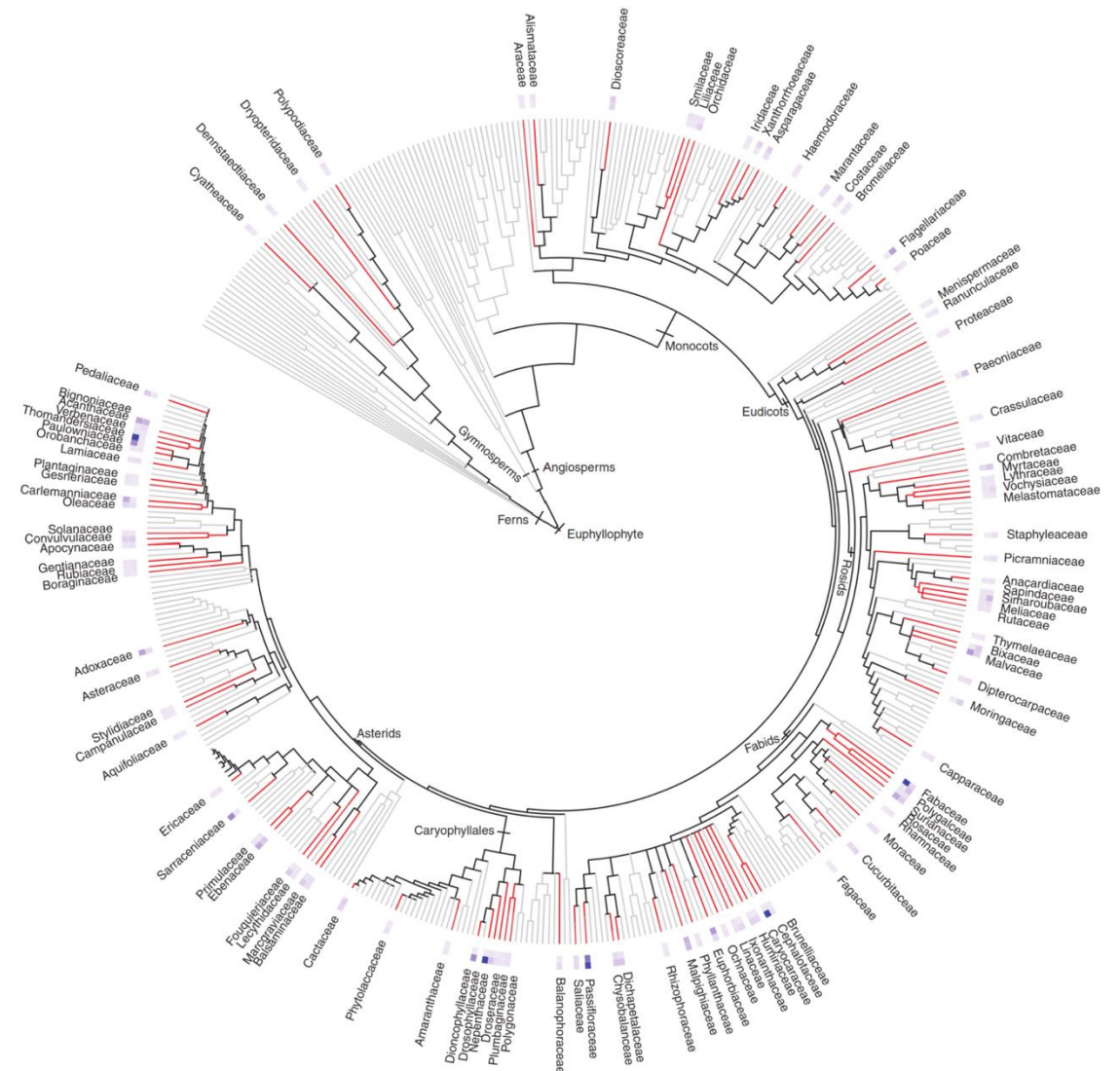


Increased flower production with
bodyguards

Rudgers, 2004

EFNs reduce herbivory and have evolved repeatedly across angiosperms.

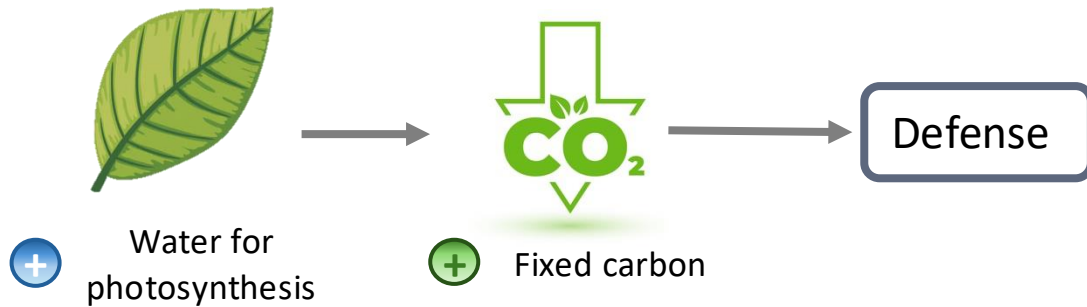
EFNs provide an ideal system to study the physiological costs of anti-herbivory defense.



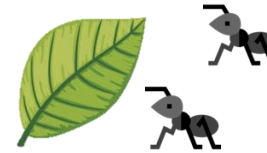
Weber & Keeler, 2013

THE COST OF DEFENDING AGAINST HERBIVORES

How can we link defense and water-use?



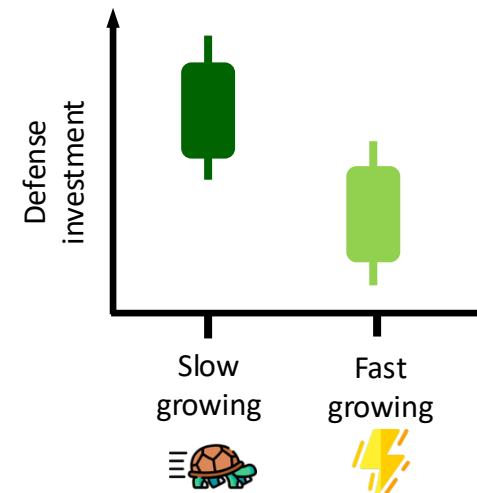
Theory in Plant Defense is HUGE!



How would this relation look like?

The Resource Availability hypothesis

Coley (1989)

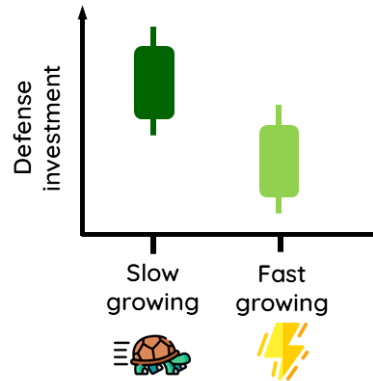


Slow-growing species should invest more in defense

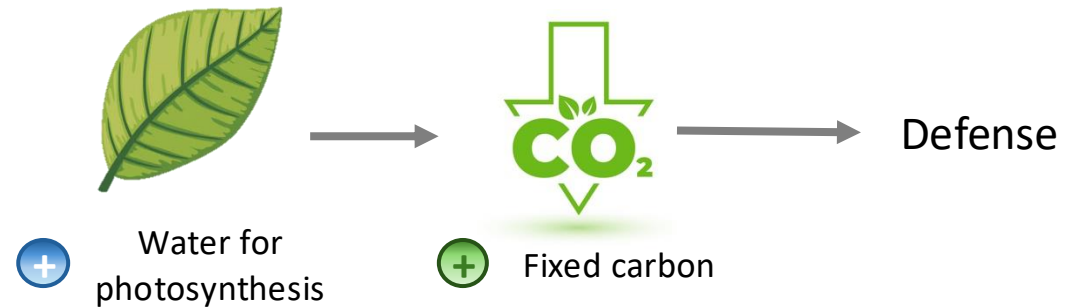
THE COST OF DEFENDING AGAINST HERBIVORES

The Resource Availability hypothesis

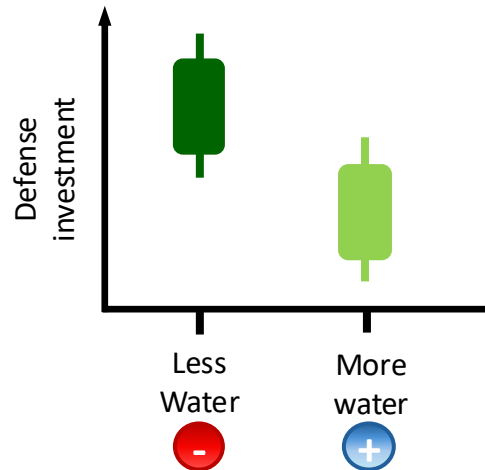
Coley (1989)



Species that **slow growth** should invest **more in defenses**



...in water vocabulary



Species with **lower water transport** should invest **more in defenses**

Cochlospermum orinocense



How do leaf water-use strategies change
with the presence of EFNs?



Albizia saman

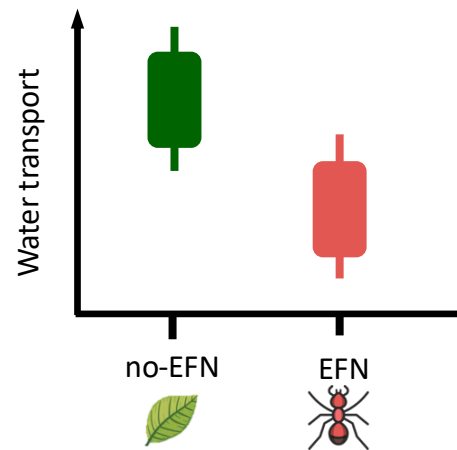
THE COST OF DEFENDING AGAINST HERBIVORES

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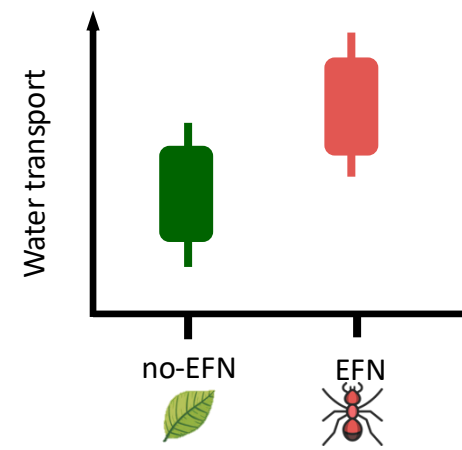
High investment in defense occurs when tissues are costly.



Species **without EFNs** will have **lower water transport** capacity

Higher EFNs efficiency

Higher investment in defense requires higher carbon fixation rates.



Species **with EFNs** will have **higher water transport** capacity than species without

Global metanalysis



Compiling a dataset with traits related to
leaf hydraulic and carbon traits



Determine the presence/absence of EFNs

Statistical analyses

Perform phylogenetically informed models
to test for the differences between groups

This study aims to provide mechanistic explanations for the repeated evolution of EFNs across lineages.

Bridging variation leaf hydraulics and defense is key to understanding **leaf ecology and evolution** of terrestrial plants

Proper data extraction and ensuring feasibility will be key challenges in the coming years.



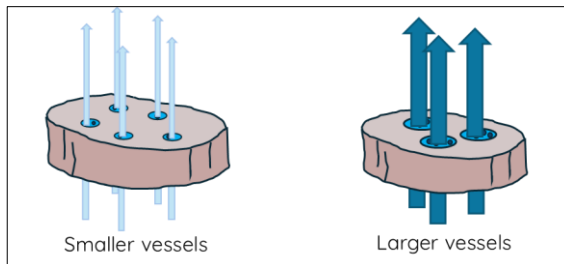
DISSERTATION PROPOSAL SUMMARY

This dissertation positions water-use strategies as an integrative dimension linking plant physiology and plant ecology.

Chapter 1

What plant–water relationships occur in tropical dry forests?

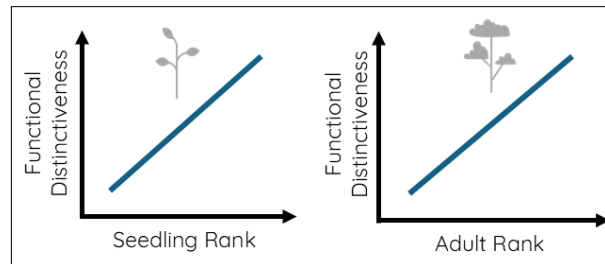
We found many strategies without a clear biophysical constraint by drought



Chapter 2

What's the role of hydraulic strategies determining species abundance?

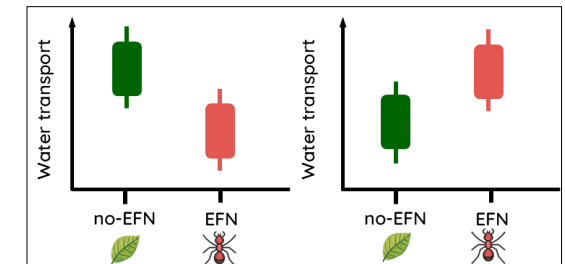
We found that common species converge towards a similar hydraulic strategy



Chapter 3

Are EFNs shaping plant hydraulics?

We aim to understand the physiological costs of plant defense



Colaboradores en Colombia Collaborations in Colombia

Dra. Beatriz Salgado
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Dra. Natalia Norden
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Dra. Camila Pizano
Dra. Camila Martínez

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Mi familia



Funding





Saguinus oedipus

Thank you!

Happy to receive any comments
or questions!

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